DIVYANSHU GOYAL

Senior Applied Scientist | Multimodal AI | Foundation Model Post-Training & Evaluation

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SUMMARY

Applied Scientist with **8+ years of experience** building **multimodal AI systems, vision-language models**, and foundation-model post-training pipelines. Specialize in supervised fine-tuning, **reinforcement learning (GRPO)**, reward modeling, **human preference optimization**, and multimodal evaluation. Lead applied research and technical development at Adobe, translating novel model architectures and training methodologies into **production AI products**. Research contributions include **first-author publications**, benchmark development, **six patent filings/applications**, and a compact photo-retouching model demonstrating **competitive human preference against frontier models**.

EXPERIENCE

Applied Scientist – Adobe Inc., San Jose, CA June 2021 – Present

GenStudio for Performance Marketing (Technical Lead)

- Led technical strategy and development of Adobe GenStudio for Performance Marketing, taking the product from **prototype to multi-million-dollar enterprise adoption** following its October 2024 launch.
- Architected the production API layer powering LLM-based capabilities for **Fortune 500** customers, enabling scalable integration of generative AI into enterprise marketing workflows.
- Co-invented automated brand-guideline extraction and compliance scoring (US20250117829), establishing the foundational engine for **Adobe Brand Intelligence**.
- Developed parameter-efficient fine-tuning pipelines on Fortune 500 customer brand data, directly enabling a **custom-model enterprise deal**.

Multimodal AI & Foundation Model Post-Training (Adobe Marketing VLM)

- Provided **technical leadership for a seven-person team** developing specialized VLMs, judge models, and AI-powered retouching capabilities.
- Optimized distributed supervised fine-tuning (SFT) and multi-round GRPO post-training for Gemma-3-12B across 8× H100 GPUs, reducing training time from **six days to 24 hours** and compute costs by **6**×.
- Designed and iterated on seven custom reward functions to improve model alignment and evaluation quality; diagnosed **reward hacking** where the policy selected middle ratings for 84% of images vs. 56% for human annotators, reducing this behavior to **64%** through reward redesign.
- Built a scalable, fault-tolerant multi-GPU annotation pipeline processing **100M+ data points in 24 hours** using a teacher VLM to generate SFT training data.
- Extended the VLM from image evaluation to image improvement by developing a **Photoshop agent** that plans and applies edits to enhance marketing appeal; integrated the model into Adobe Photoshop's agent experience, used by **millions of customers globally**.

ATune: Unsupervised Photo Retouching (Co-Lead with Adobe Research)

- Co-led research and system architecture as **co-first author**, owning the synthetic data pipeline, model training recipe, and human evaluation framework; mentored a PhD intern.
- Developed a compact **54M-parameter model** that performs non-destructive photo adjustments through off-the-shelf photo editors, learning from **10M unedited photographs** without expert-generated targets.
- Evaluated against frontier models in a 157-rater human preference study (3,524 judgments), achieving **54.3% of decisive votes against GPT-6 Astra** (95% CI 50.3–58.3) and **matching Claude Opus 5** (50.5%).
- Demonstrated **data scaling**, with model quality improving as the training corpus increased from 1M to 10M unlabeled photographs, reducing dependence on expensive expert annotations.

JEV-Style Marketing Judge Model (Human Preference & Model Evaluation)

- Fine-tuned a Qwen3.5-4B model on 1.1M human-rated examples from 14 sources, improving agreement with human image-quality assessments on KonIQ from 0.421 to **0.663 SRCC (+57%)**.
- Identified customer pain points and translated judge-model capabilities into product demonstrations, driving **integrations with Adobe Lightroom, Adobe Stock, and Adobe Firefly AI Assistant**.
- Diagnosed vision-encoder collapse that made training checkpoints image-blind within 200 steps despite apparently normal loss curves; implemented automated **vision-health monitoring** to detect and halt silent training failures.

Evaluation Research (DistortBench)

- Developed **DistortBench** (**first author**), a 13.5K-question benchmark for fine-grained visual quality evaluation across 27 distortion types and five severity levels.
- Established a systematic evaluation framework for assessing VLM capabilities in visual distortion identification; **adopted internally at Adobe** to evaluate models powering the Photoshop agent.

Machine Learning Engineer – Adobe Inc., Bangalore, India July 2016 – August 2019

- Joined Adobe's core ML serving platform team at inception, helping architect end-to-end model-serving infrastructure for internal ML workloads, including field-aware factorization machines.
- Scaled production inference throughput to **3,500 QPS per node at 0.3–0.6 ms p99 latency**, implementing a garbage-collection-free asynchronous event pipeline using LMAX Disruptor (featured on the Adobe Tech Blog).
- Developed model lifecycle infrastructure supporting versioned deployment, monitoring, and staged production rollouts.

PUBLICATIONS

ATune: Unsupervised Photo Retouching Through Image-Editor Controls
D. Goyal*, G. Ruiz*, A. Meisami | *Under review* | *Equal contribution

DistortBench: Benchmarking Vision Language Models on Image Distortion Identification
D. Goyal, A. Eppa, V. Bannihatti Kumar | *CVPR & ICML 2026 Workshops* | arXiv:2604.19966 | First author

JudgeBreak: Is Taste Hidden in the Layers? Stress-Testing Reward Models and LLM Judges with Representation-Based Subjective Evaluation
V. Bannihatti Kumar, A. Narang, M. Anand, A. Eppa, **D. Goyal** | *ICML 2026 Workshop* | OpenReview

Curiosity-Driven LLM-as-a-Judge for Personalized Creative Judgment
V. Bannihatti Kumar, **D. Goyal**, A. Eppa, N. Bhandari | *AAAI 2026 Workshop (**Oral**)* | arXiv:2510.05135

PATENTS & PATENT APPLICATIONS

Title	Number	Year
Guiding Language Translation with Translation Documents Using ML	US20260050753	2026
Brand-Aligned Marketing Content Generation (Adobe Brand Intelligence)	US20250117829	2025
Cultural Adaptation of Images Using Generative AI	US20260051084	2026
Parameter-Efficient Fine-Tuned Models	US20250217193	2025
Video-to-Video Culturalization Using Generative AI	Filed with USPTO	2026
Learning to Be Surprised: A Curiosity-Driven Framework for Creative Judgment	Filed with USPTO	2026

SELECTED INDEPENDENT PROJECTS & TECHNICAL CONTRIBUTIONS

VibeNanoChat: GPT-2-Scale LLM Pretraining from Scratch 2026 | code
Pretrained GPT-2-scale transformers from scratch at **over 1M tokens/sec** on 8× H100 GPUs using a custom distributed Muon optimizer. Derived empirical compute-optimal scaling frontiers through isoFLOP profiling across 40+ training runs. Published technical blogs: Going Fast, Scaling Transformers.

BrownBearAI: On-Device Photo Understanding for iOS 2026 | App Store
Built and shipped a privacy-first iOS application for on-device neural face clustering and deduplication, achieving **99.4% BCubed F1** on LFW identities with zero cloud dependency.

LaSeRo: Antarctic Rock Outcrop Segmentation 2019–2020 | code
Trained and evaluated SegNet architectures for geological rock-outcrop segmentation across multispectral Landsat 8 satellite imagery.

Face Mask Detection 2020 | code
Developed a real-time mask detection pipeline, benchmarking CNN architectures with VAE-based representation learning and Grad-CAM visual attribution.

TECHNICAL SKILLS

Applied AI & Foundation Models: Multimodal LLMs, Vision-Language Models (VLMs), Supervised Fine-Tuning (SFT), GRPO, RL Post-Training, LoRA/PEFT, Reward Design, LLM-as-a-Judge, Reward Models, Model Alignment

Model Evaluation & Research: Human Preference Modeling, Image Quality Assessment, Benchmark Design, Subjective Evaluation, Representation Analysis, Synthetic Data Generation, Scaling Laws, Human Evaluation Studies

Distributed ML & Infrastructure: PyTorch, vLLM, Distributed Training (DDP, ZeRO-2), Multi-GPU Training, FlashAttention-3, torch.compile, Liger Kernels, Muon, Mixed Precision (BF16)

Development & Tooling: Python, Hugging Face Transformers, Hugging Face PEFT, Weights & Biases

EDUCATION

Institution	Degree	GPA	Years
Georgia Institute of Technology	M.S. Computer Science (Machine Learning)	4.0/4.0	2019 – 2021
BITS Pilani	B.E. Computer Science, M.Sc. Mathematics	8.75/10	2011 – 2016